

Building the AgiMo Digital Twin

Towards continuous transport model assimilation



Data-driven agile planning
for responsible mobility

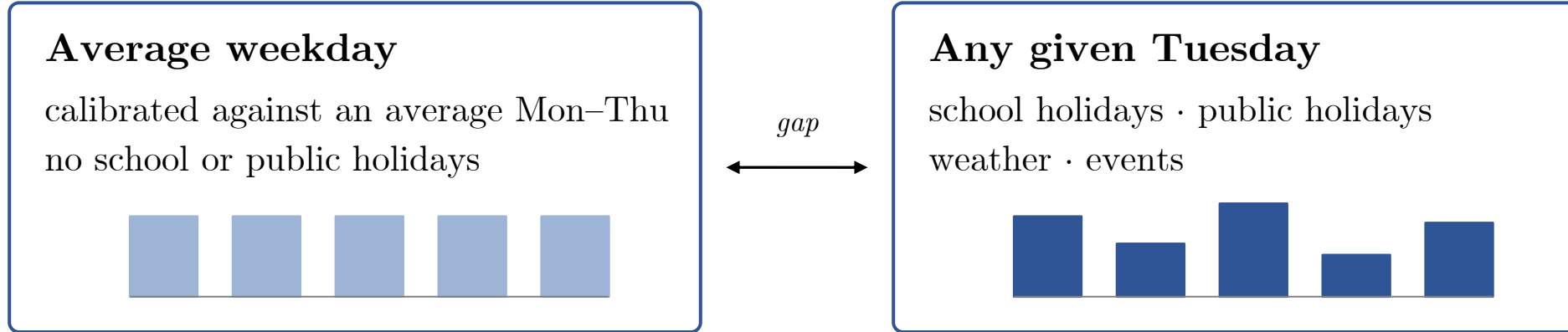
SFB
TRR 408



Daniel Röder
VSP - TU Berlin

MATSim User Meeting, Paris, September 2026

A calibrated scenario represents an average day — not yesterday



Research objectives

“Which data and methodologies need to be combined [...] to build a credible planning digital twin [...]?”*

- **Is model assimilation with mobilephone data possible?**
- Is automated data-feed deployment feasible?
- Is daily automatic calibration possible?
- Can the computational speed be improved?

Digital twin vs. Digital shadow

Digital model



Digital shadow



Digital twin



————> automatic data flow

- - - -> manual data flow

AgiMo implementation

The open question is not the degree of
automation —

it is whether the incoming data carries

enough signal to move the model.

Base scenario and reference data

Base scenario

matsim-berlin, open scenario

- 1 pct sample
- calibrated against 2018 counts, approx. 400 stations
- based on release v7.1.1
- freight left unchanged

Reference counts

Berlin traffic detectors

- 2023 counts, 185–245 stations
- Hourly values
- not all counts carry data for every day/hour
 - data-cleaning is very important

Keep in mind:

- Calibrated against 2018, but compared against 2023
=> model will be updated in the future
- Throughout this talk we therefore analyze only the **difference between baseline run and assimilated run** on the same day

12 tuesdays in 2023, different conditions

Why historic days?

More complete data, no risk of broken APIs, compare across model versions.

Why 2023?

First year after the pandemic with stable travel behaviour, SrV was conducted.

Why Tuesdays only?

Removes the weekday effect, remaining variation is seasonal, weather and holidays.

Why these twelve?

Spread over the year to cover different weather, school holidays in and out, and one public holiday.

date	wind [m/s]	rain [mm]	clouds [%]	temp. [°C]	school holiday	public holiday
2023-01-31	6.9	5.6	75	4.3	yes	no
2023-02-07	0.9	0	10	-2.7	no	no
2023-02-28	1.7	0	15	1.2	no	no
2023-05-30	3	0	15	16.7	yes	no
2023-07-11	2.5	0	65	24.1	no	no
2023-07-18	3.2	0	40	19.7	yes	no
2023-08-29	2.5	5.5	100	15.4	no	no
2023-09-05	2.6	0	40	21.5	no	no
2023-09-26	1.9	0	45	17.4	no	no
2023-10-03	4.6	9.1	80	18.1	no	yes
2023-10-24	2.5	12.7	95	11.4	yes	no
2023-12-05	4.5	1.6	100	-1.2	no	no

Daily indicators from mobile phone data

1 Source

- Zip-code level, per calendar day
- Covers the full scenario area
- Telefónica, scaled to full population

2 Indicators

out-of-home rate

trips per mobile person

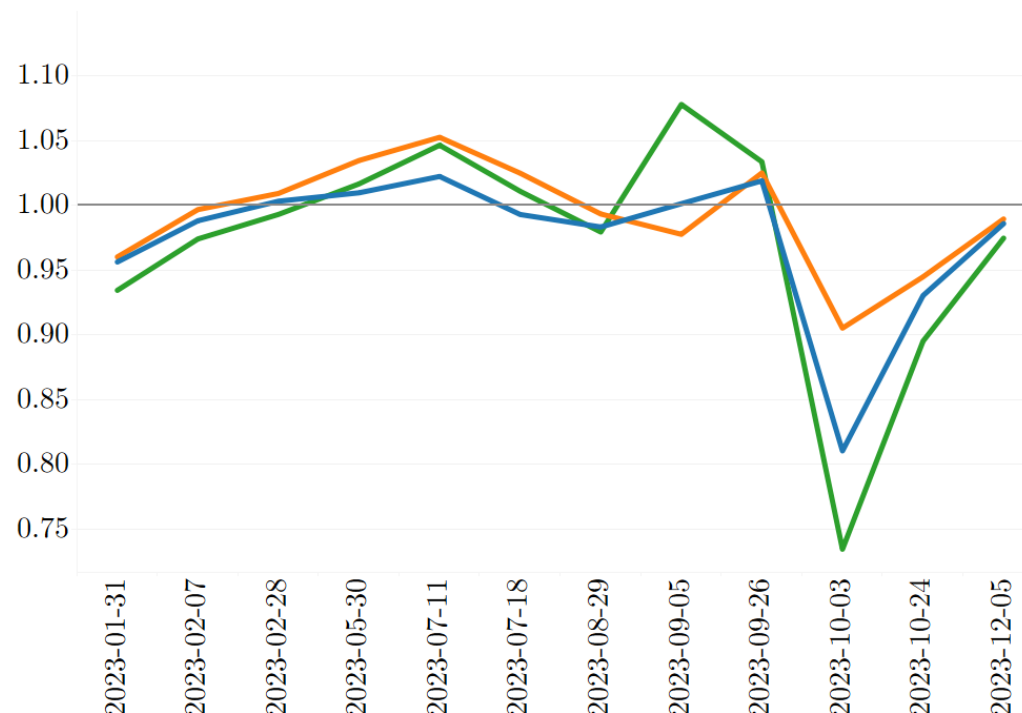
out-of-home duration share

3 Baseline

$\text{baseline (d)} = \text{value (d)} / \text{reference}$

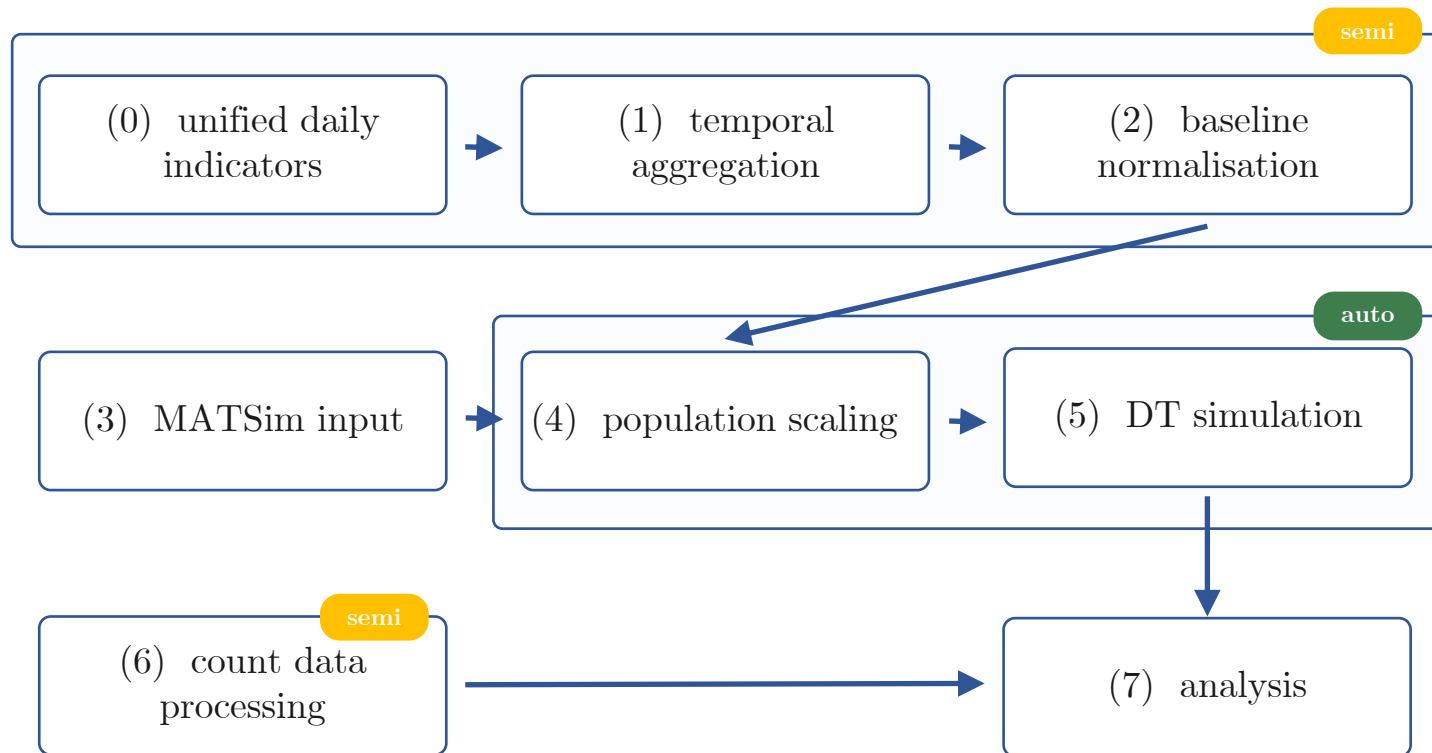
reference = median over all Mon–Thu outside school
and public holidays, per zip code

baselines, averaged over all zip codes · 2023



Almost all days are within a range of $\pm 5\%$

The assimilation pipeline



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Population scaling, but how?

Starting point: scenario with out-of-home rate < 1 — no plans exist for stay-home agents

How many agents change?

sets the out-of-home rate

baseline < 1 Mobile agents stay at home

- Cheap — no new destinations needed
- Remove plans from selected mobile agents

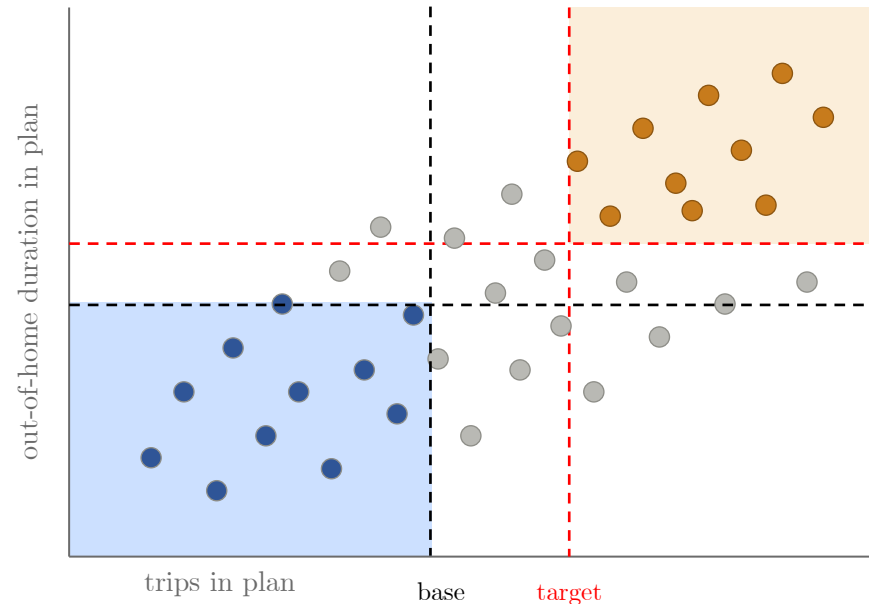
baseline > 1 Stay-home agents need a plan

- Expensive — location choice needed, no data to base it on
- Instead: clone a neighbouring agent, reuse its destinations

Freight is left as-is in both cases.

Which agents change and how?

the plan must pull trips per person and out-of-home duration toward the target



baseline < 1 : remove from this set · **baseline > 1 : clone from this set**

Both criteria must hold at once, so only a corner of the population qualifies — and cloning adds a third filter: the donor must be a neighbour.

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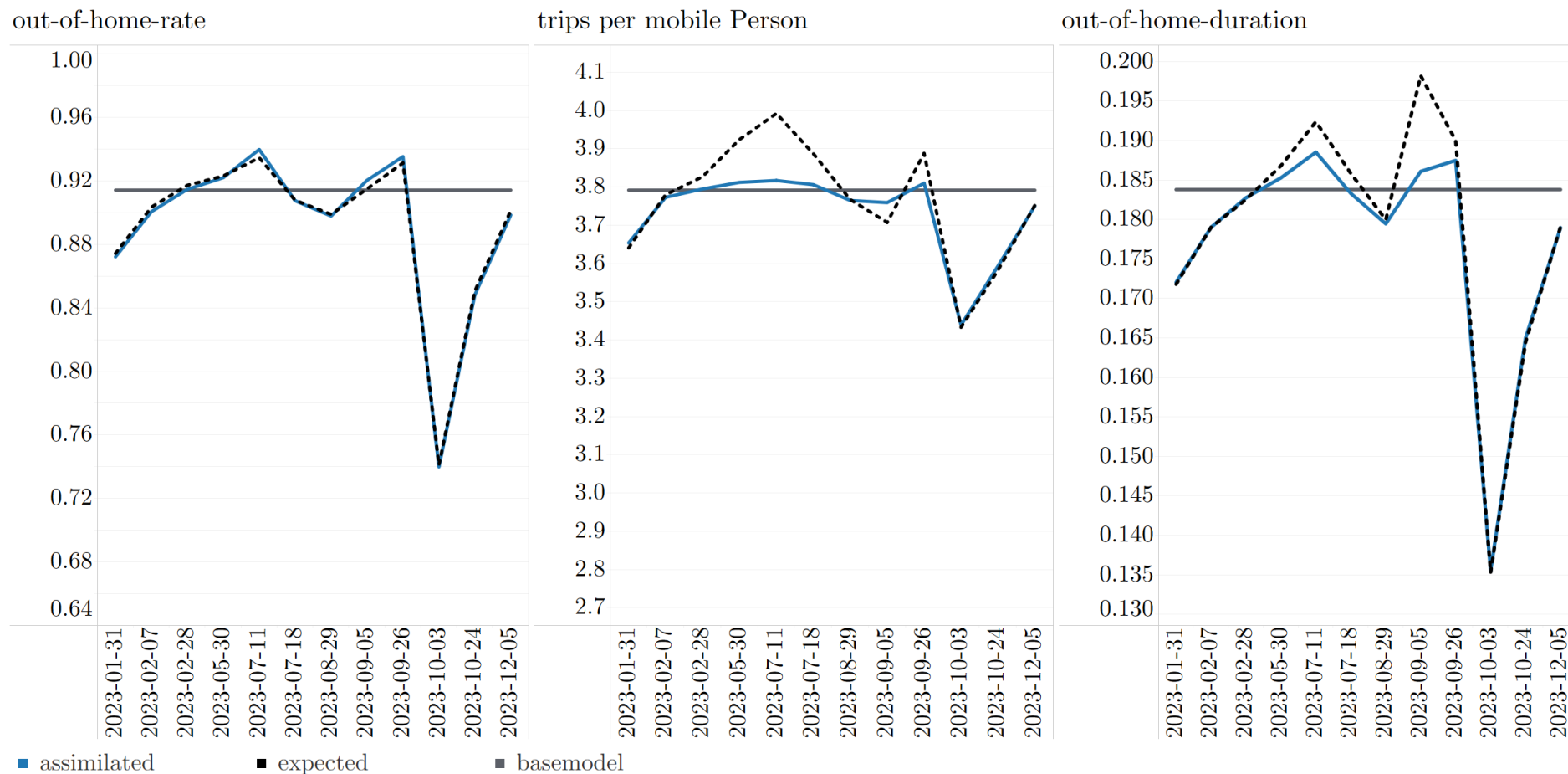
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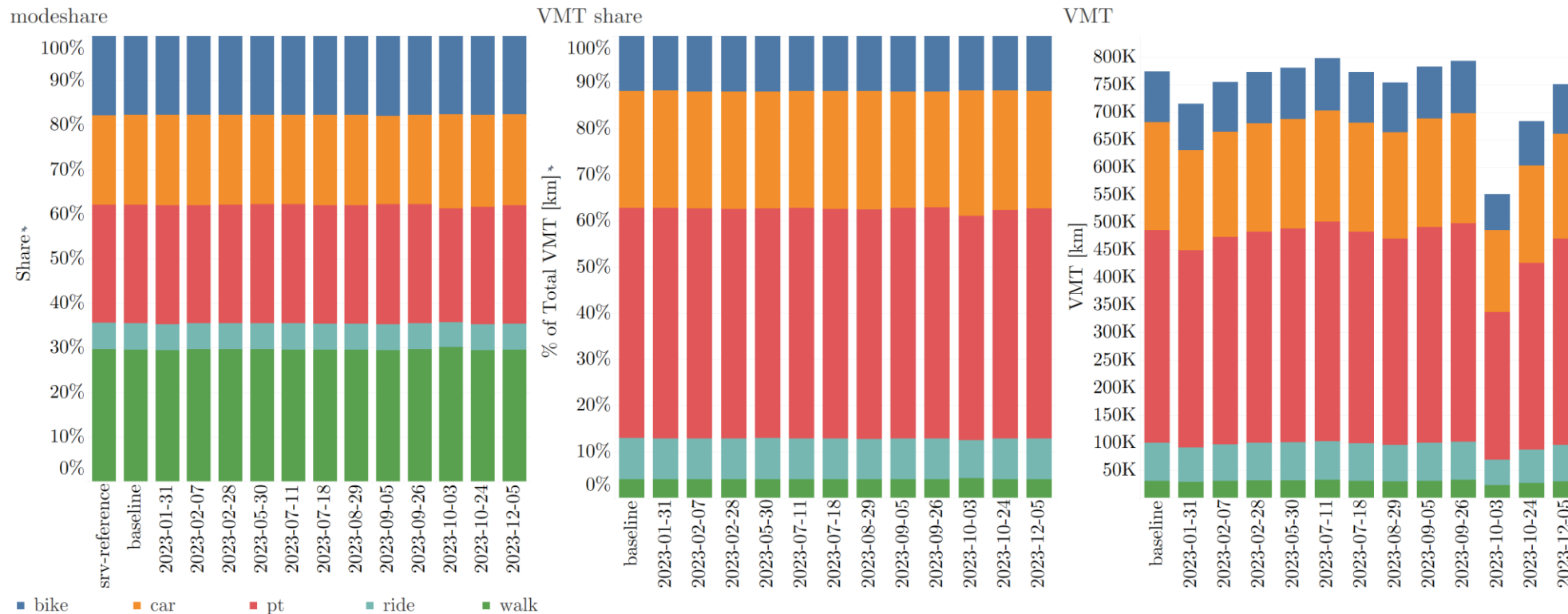
Reductions work, increases do not



- Good fit for reductions
- Increased baseline (trip and duration) does not move
- Assumption: Sampling of the donor-plans is too restrictive

Activity-level says nothing about modes

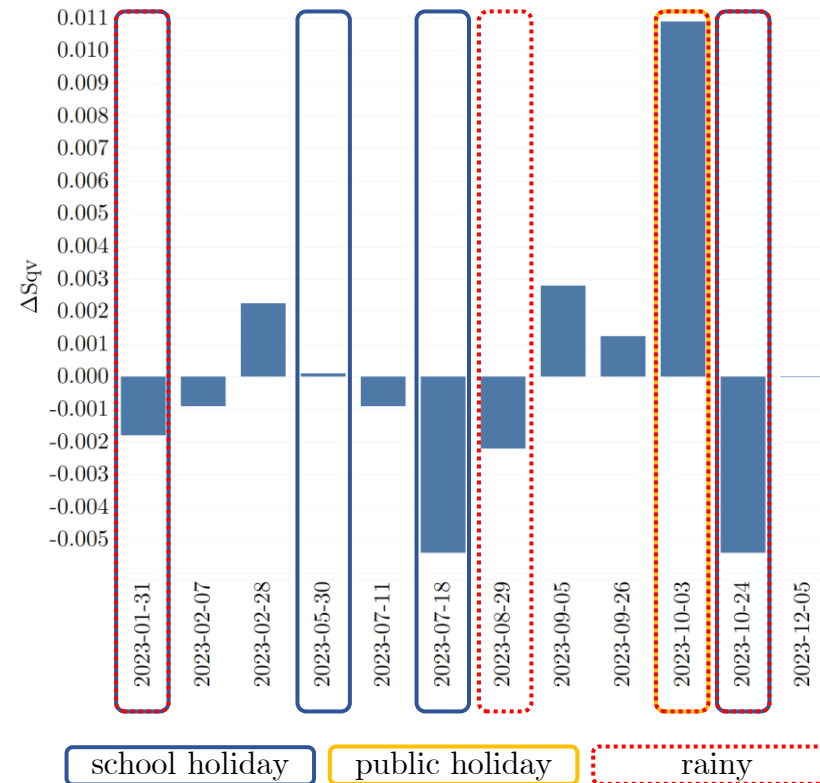
- Mode share stays nearly unchanged
 - Even Oct 3rd with the highest effect on indicators is nearly unchanged
- VMT changes in level, but the split between modes does not



Link counts barely respond

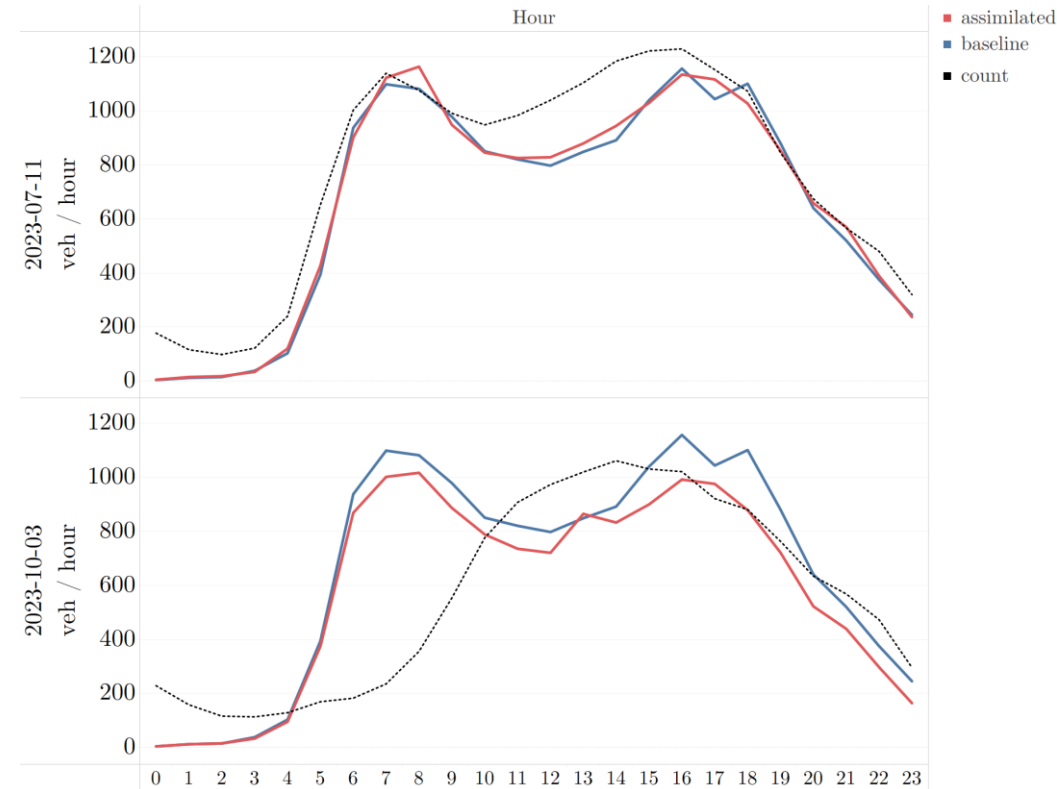
- Overall very small changes
- Biggest effect on Oct 3rd
- Negative effects on rainy days
 - systematic?
 - But 2023-07-18 is also negative

ΔS_{qv} per link and hour, assimilated - baseline, median over all measures

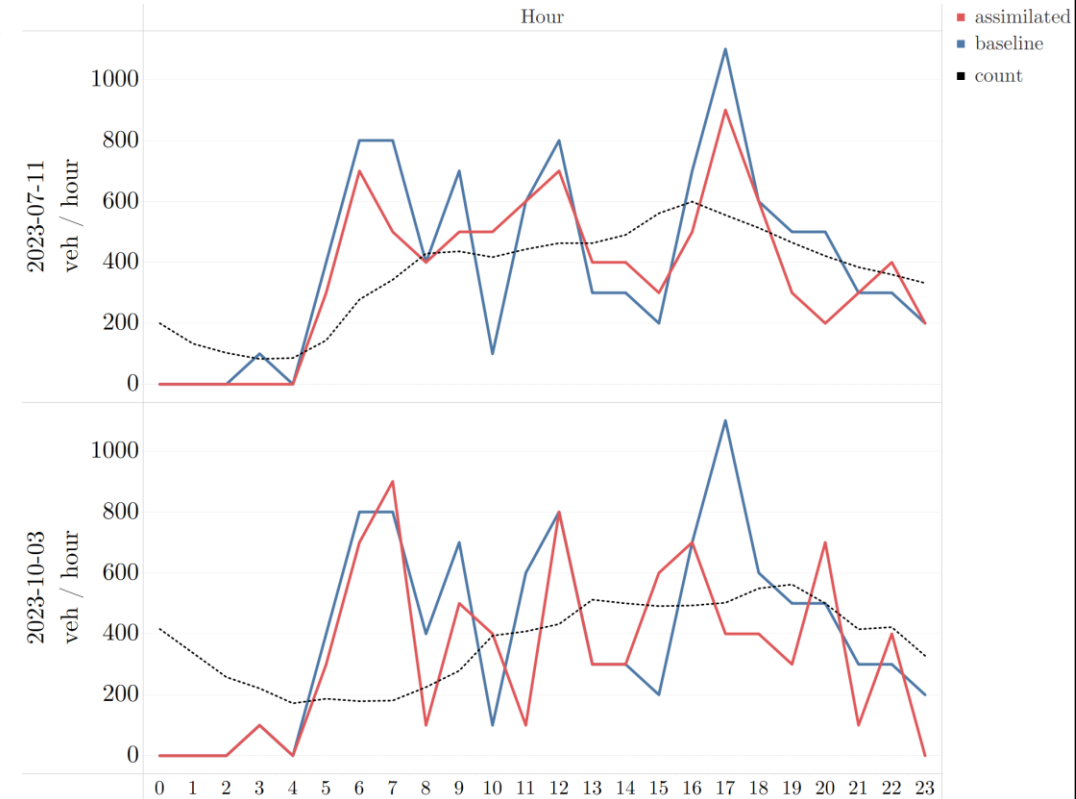


Scaling moves the level, not the shape of the day

avg over all links



single link



Conclusion

automated & regular assimilation is feasible

pipeline runs partly automated; population follows the indicators where they point down

Activity indicators carry no mode and no time structure

mode share does not move; the shape of the day stays a weekday — more agents will not fix this

Day-to-day variation is ± 5 % - possibly below the noise of a 1 % sample

only Oct 3rd, at ~ 20 %, produces a visible effect in the counts

Upscaling does not work well yet

donor-plan sampling too restrictive; affects increases only, reductions are fine

The logo for AgiMo, featuring the text "AgiMo" in a serif font, with the "i" in "Agi" and the "o" in "Mo" having dots. The logo is set against a blue grid background.

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Outlook



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next

Evaluate signal-noise-ratio

Run different 1%-samples. Run 10%-sample.

next

Update matsim-berlin

Update counts to 2023. Pull in recent changes from the pipeline (Activity-Time-Structure, fully automated build)

next

Improve modeshare assimilation

We assume modeshare is affected by season & weather. Pull in further data: bike counts, “Stadtradeln”, weather

next

Update infrastructure

Currently network and transitschedule are constant. We plan to incorporate regular updates.

later

Move to livefeeds

later

Automatic calibration & computational speed

Thank you!



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Appendix: State of the art

ODTP

ETH Zürich

SCOPE

Orchestration framework for coupling simulation components

MODEL

Connects to MATSim / Equisim — agent-based

STATUS

No development since Feb 2025

A framework, not a running twin of a city.

ONE

Transport for London

SCOPE

Greater London road network: 53,500 km, 17,000 junctions, 5,500 signalised

MODEL

PTV Visum — macroscopic, network-focused

STATUS

In operational use for scheme appraisal

Not activity-based, no per-day assimilation.

Digitaler Zwilling

Landeshauptstadt München

SCOPE

City-wide platform: 3D city model, sensors, construction sites, AR/VR

MODEL

Infrastructure and visualisation layers; mobility is one of many

STATUS

Funded project, active

No calibrated travel demand model underneath.

None of them updates a calibrated, activity-based demand model to a specific calendar day from running data feeds — the AgiMo Digital Twin starts here!

The AgiMo logo is displayed within a blue grid pattern. The text 'AgiMo' is in a black serif font, with the 'i' being lowercase and the 'A', 'g', 'M', and 'o' being uppercase.

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Appendix: SQV distributions



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